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ARTICLE

Soft and hard tissue changes following immediate implant placement and immediate loading in aesthetic zone—a systematic review and meta-analysis

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KEY POINTS

- Evidence from numerous studies have reported that immediate placement of implants into fresh extraction sockets and have demonstrated predictable results.
- The current evidence highlights placing dental implants directly into recently extracted sockets could help preserve the shape of the ridge. This may help retain some bone volume, potentially leading to better soft tissue aesthetics.
- This systematic review provides assessment of the soft and hard tissue changes following immediate implant placement and loading in maxillary aesthetic zone, which may help the implantologists in better understanding and prediction of the aesthetic outcomes.

BACKGROUND: Type I immediate implant placement has its own advantages like reduced treatment time, number of surgeries and post-extraction bone loss, however, the presence of insufficiently keratinized mucosa poses a challenge for flap adaptation and hinders the achievement of primary stability. Additionally, scientific evidence supports the notion that post-extraction bone loss is a natural biological occurrence that can impact the success of treatments.
OBJECTIVES: The primary outcome was to find out the hard and soft tissues changes around the implant following immediate placement and immediate loading. The secondary outcome was to record the adverse events post implant placement such as infection during the course of healing and after restoration, implant failure which would include surgical and post restoration, and over- all success and survival rate of implant.
MATERIAL AND METHODS: A detailed electronic literature search of the articles published in English language was undertaken in October 2023 on online search engines Medline/PubMed and Cochrane databases with no restriction on year of search to include studies on immediate implant placement and loading with a mean follow-up time of at least 1 year. Weighted means of soft and hard tissue changes were obtained by the inverse variance method.
RESULTS: A total of 16 studies were included. There was no difference in crestal bone levels in immediate or delayed implant placement with immediate provisionalization in the anterior maxilla. The peri-implant margin remained and no differences in papillary loss was seen when compared to the delayed implant placement cases. Papilla were more stable or showed less recession in flapless approach compared to full thickness flap approach. Among the studies which filled the GAP with bone graft materials, no significant changes were found in the bone level changes. In case of recession, immediate implant placement with provisionalization did result in approximately 1 mm less facial gingival recession compared with that in the group that had a socket graft. Implant related complications occurred more in immediate implant placement and provisionalization compared to delayed group. And almost similar implant success and survival rates were seen in comparison to delayed implant placement groups.
CONCLUSIONS: Despite the similar implant survival rates observed in comparison to delayed implant placement groups, more long-term studies are necessary to determine the success of immediate implant placement and immediate loading. Special attention has to be given to aesthetic outcomes.

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INTRODUCTION

The implant is strategically positioned within the socket immediately after the tooth extraction, allowing for the seamless replacement of the extracted tooth in a single surgical session, thereby introducing the innovative concept of immediate post-extraction implants[1].

Immediate implant placement offers numerous benefits, including a decrease in the number of surgeries and overall treatment time. Additionally, patients tend to have better acceptance of the procedure due to the psychological advantage of promptly replacing a missing tooth with an implant. Another advantage is the ability to position the fixture ideally along the

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axis, as well as preserving the height and width of the alveolar ridge, resulting in aesthetic benefits[2].

After tooth loss, a number of biological processes occur: vertical and horizontal bone reabsorption, resulting in changes to the height and thickness of the alveolar bone; gingival collapse; movement of neighbouring teeth; and changes in bone quality, including compact bone collapse and alveolar bone marrow formation[1].

The period between tooth extraction and implant placement is when most bone loss and gingival remodelling occur, often causing biological, aesthetic, and functional challenges. After this remodelling phase, the alveolus may no longer have a sufficient width for implant placement, which can sometimes limit treatment options[3].

Studies have shown that placing dental implants directly into recently extracted sockets could help preserve the shape of the ridge. This may help retain some bone volume, potentially leading to better soft tissue aesthetics[4–6]. As a result of these various factors, the concept of placing implants right after tooth extraction was suggested as a method to preserve the bone structure in the surgical site. Numerous studies have reported on immediate placement of implants into fresh extraction sockets and have demonstrated predictable results for this treatment approach[7–10].

An increasing volume of data suggests that a treatment plan involving the immediate extraction of a tooth, followed by the placement of an implant and loading, can be effectively implemented in patients with hopeless maxillary anterior teeth[11–15] Numerous clinical studies have demonstrated that immediate implants and immediate provisionalization procedures can lead to high success rates and implant survival. Nevertheless, a growing body of literature is highlighting the potential aesthetic drawbacks of this approach, such as the risk of buccal gingival recession and labial bone volume loss in the long term[16–21].

Hence the aim of this systematic review was to assess the soft and hard tissue changes following immediate implant placement and loading in maxillary aesthetic zone.

MATERIALS & METHODS

This systematic review was based on PRISMA (preferred reporting items for systematic reviews and meta-analyses) guidelines[22] (PROSPERO ID: CRD42022308780).

Focused question

The PICO (population, intervention, comparison and outcome) strategy was used; the question was—What are the soft and hard tissue changes (O) seen following immediate implant placement and loading (I) in maxillary aesthetic zone (P) compared to early and/or delayed implant placement and loading (C)?

Inclusion criteria

Randomised controlled clinical trials (RCT), controlled clinical trials (CCT), prospective studies, retrospective studies related to immediate implant placement and loading (restorations delivered within 48 h of implant placement) in the maxillary aesthetic zone (including incisors, canines and premolars) reporting on primary outcomes of hard and soft tissue changes following immediate loading with a mean follow-up of 1 year or more; studies with a minimum of 10 or more patients examined clinically and followed-up, studies reporting number of implants placed per subject, describing the three-dimensional positioning of the implant and restorative protocol; and studies reporting additional outcomes like number of failed implants; and articles published in English were included.

Exclusion criteria

Excluded from the analysis were animal research, laboratory studies, quantitative analysis, research centred on patient outcomes, as well as studies utilizing patient records, surveys, questionnaires, or interviews, along with reviews and case studies

Outcome measure

The primary outcome was to find out the hard and soft tissues changes following immediate implant placement and immediate loading. The secondary outcome was to find the adverse events such as infection, implant failure, success and survival rate of implant.

Search strategies

In October 2023, an extensive electronic literature review was conducted on Medline/PubMed and Cochrane databases to search for the article published in English language, without any limitations on the year of search. (Supplementary Table 1). Additionally, a manual search was conducted to find other suitable studies by examining the references of the included studies and other published reviews. Implant journals that were searched included International Journal of Oral and Maxillofacial Implants, Clinical Oral Implants Research, Journal of Oral Implantology, Implant Dentistry, Journal of Periodontology, Clinical Implant Dentistry and Related Research and European Journal of Oral Implantology.

Selection of studies

Studies were selected firstly by removing the duplicate articles, then screening the titles and abstracts of the remaining articles for eligibility. After applying inclusion and exclusion criteria, the complete texts of eligible articles were read. Two reviewers (RG and KS) conducted and completed the article selection in accordance with delineated inclusion and exclusion criteria. A third reviewer (SM) re-assessed the article selection strategy and was further consulted when uncertainties arose on whether to include or exclude articles. The reviewers were blinded, and the assessments were conducted in parallel. Discrepancies were resolved through an established consensus between reviewers. A fourth reviewer (RC) was consulted if consensus was not reached between reviewers. The level of agreement among the reviewers was assessed using Kappa statistics.

Data extraction

Data was extracted from each study under the following headings —author, year, study design, centre, number of patients, gender, smoking/systemic diseases, reasons for extraction, number of implants placed (D=diameter; L=length in mm), implant position (apicocoronal), implant stability assessment, insertion torque and/ISQ, gingival biotype, flap/flapless technique, graft or any other material or growth factors used, pre/post operative antibiotics prescribed, type of definitive prosthesis, retention type and delivery time, patients and implants lost during follow-up, follow-up time, number of implants failed, success and survival rate of loaded implants. Soft and hard tissue outcomes included measurement method of soft tissue changes, probing depth (>3 mm), mid-facial recession (mm), papillary recession (mm), papillary index if any used like Pink aesthetic score (PES), measurement method of hard tissue changes and bone loss (mm or %).

Quality assessment

The quality of study and risk of bias was assessed independently by two reviewers (R.G. and R.C). The Newcastle-Ottawa scale (NOS) was utilized to evaluate the quality of case-control and cohort studies[23], with a threshold of six or more points indicating high quality. The Cochrane collaboration's tool was employed to assess the risk of bias in RCTs, focusing on sequence generation, allocation concealment, incomplete outcome data, and blinding. Studies were categorized as having low, unclear, or high risk of bias based on the inclusion of these criteria[24].

Data synthesis and meta-analysis

Meta-analysis was performed using statistical R software (R Core Team, 2018 version 3.4). Papillary loss, bone loss and implant failure were the outcomes measured. Papillary loss, bone loss expressed in mean differences and implant loss expressed in risk estimates and 95% confidence intervals (CIs). A fixed-effects (FE) model was used where statistically insignificant heterogeneity was detected. The implant served as the statistical unit for 'implant failure. The I^2 statistic was used to demonstrate the heterogeneity of the studies, with 25% indicating low

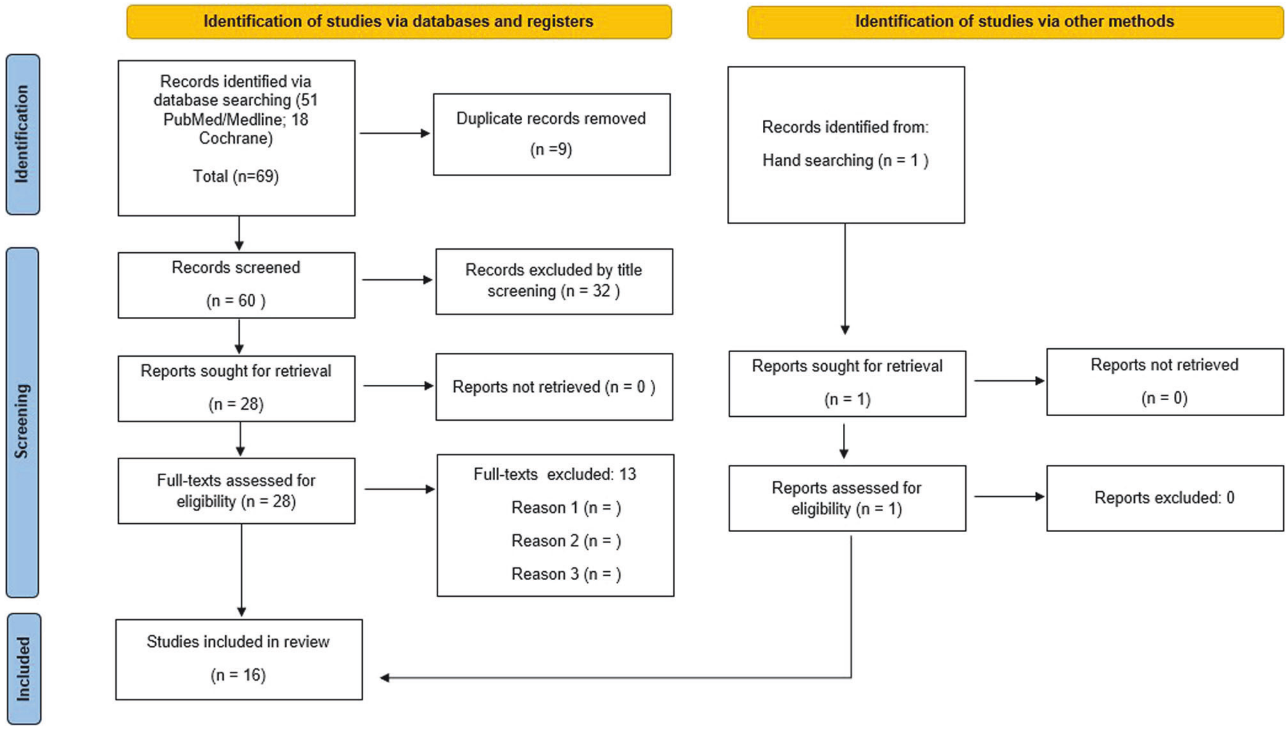


Fig. 1 PRISMA flowchart.

heterogeneity, 50% indicating moderate heterogeneity, and 75% indicating high heterogeneity. A random-effects meta-analysis was applied where suitable, following an assessment of heterogeneity and consideration of variations in treatment approaches, populations, and study environments. Mean differences were used to present pooled estimates, in line with the predefined outcome characteristics, accompanied by 95% confidence intervals (CIs) as measures of precision. To assess statistical heterogeneity, an I-squared test was conducted, where a p-value of less than 0.10 ($p < 0.10$) signalled non-homogeneity. To assess heterogeneity, confidence interval overlap was visually inspected within forest plots. The between-study variance was quantified as tau-squared (τ^2). When data were missing, the original study authors were contacted to request further information. In split-mouth designs, the reduced variability between quadrants was accounted for by applying a formula to approximate the standard deviation of the difference:

$$SD_{diff} = \sqrt{sd_1^2 + sd_2^2 - 2rsd_1sd_2}$$

setting the correlation coefficient "r" at 0.5 for split-mouth designs (with $r = 0$ for parallel designs). Publication bias was planned to be evaluated using standard funnel plots and Egger's regression test.

Publication bias

A funnel plot was constructed to detect publication bias. The presence of publication bias was suggested by an asymmetry in the funnel plot. The classic fail-safe N test was employed to measure the extent of publication bias. Additionally, both a rank correlation test and a regression test were conducted to assess funnel plot asymmetry.

RESULTS

Literature search

Initially, the literature search resulted in 69 papers. Sixty papers remained after excluding duplicate articles, out of which 32 papers were further excluded following the screening the abstracts, resulting in 28 studies. The hand search resulted in one additional paper. Thirteen studies did not fulfil the inclusion criteria, hence were excluded (Fig. 1), and finally, 16 articles[3,25–39] were included in this review. The reviewers had a high level of agreement ($k = 0.856$).

Study description

An overview of the 16[3,25–39] included papers (5 prospective[3,25–28], 2 retrospective[29,30] and 9 RCT[31–39]) is tabulated (Supplementary Table 2). Six studies[28,31,32,34,35,39] were conducted at universities, 2 at private clinics[29,37], 6 were multicentre studies[3,26,27,30,33,36], 1 at a hospital[38] and 1 study did not mention the centre[25]. Total number of patients included in 16 studies were 513, minimum number of patients were 16 in a study by Noelken et al.[27] and the maximum were 55 in a study by Cooper et al[3]. Number of smokers were 42 reported by 7 studies[26–28,34,36,37,39], one was controlled diabetes reported by Grandi et al.[26] Total number of implants placed in 16 studies were 523. Implant dimensions were reported by 14 studies[3,25–27,29–33,35–39] and implant position (apicocoronal) was reported by 12 studies[25,26,28,29,31–38]. Implant stability assessment was done on 14 studies[3,25–28,30–38]. Five studies[29,30,33,35,39] reported on gingival biotype, 10 studies[26–28,30–35,38] used flapless technique, 3 studies[29,36,37] used flap technique and 3 studies[3,25,39] reported using both flap and flapless techniques. Grafting was used 7 studies[26,27,33–37] and one study[25] reported using graft along with membrane. Ten studies[25–27,30,31,34–38] reported on prescribing both pre and post operative antibiotics, one study[33] reported on prescribing pre-operative antibiotics only and two studies[29,32] reported on prescribing post-operative antibiotics. Thirteen studies[3,26–34,36–38] reported on the type of definitive prosthesis delivered, of which 7 studies[26,29,31–33,36,37] reported on delivering metal-ceramic crowns, 3 studies[3,28,30] delivered all ceramic crowns, 2 studies[27,38] delivered zirconia crowns and one study[34] reported on delivering both all-ceramic and metal-ceramic crowns. The type of retention for definitive prosthesis was cement-retained in 12 studies[3,25,27,29–32,34–38] and 3 studies[26,28,33] used screw-retained type, all of which were delivered by 2 to 6 months. Total of 50 patients and 36 implants were lost in follow-up. Total of 20 implants failure were reported by 10 studies[3,26–28,30,32,34,36,37,39]. Implant success rate was reported by 8 studies[25–29,33,34], 100% was reported by 6 studies[25,28,29,33–35] and least was found to be 90% reported by Noelken et al[27]. Implant survival rate was reported by 7 studies[3,27,29–31,35,38], 100% was reported by 4 studies[29,31,35,38] and least was 94.55% reported by Cooper et al.[3]

Soft and hard tissue changes following immediate implant placement and loading are given in Table 1. Probing depth of more than 3 mm was reported by Spinato et al.[35] Mid-facial recession was reported by 9 studies[3,26,28–30,34,35,38,39] and 7 studies[3,26,29,30,34,35,39] reported on papillary recession. Papillary index by Jemt was used by 3 studies[26,28,32] and PES by

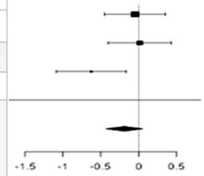
Table 1. Overview of soft and hard tissue of the included articles.

Reference	Measurement method (soft tissue)	Probing depth (>3mm)	Mid-facial recession (mm)	Papillary recession (mm)	Papillary index used	Pink esthetic score(PES)/ white esthetic score (WES)	Measurement method (hard tissue)	Bone loss (mm or %)
Di Alberti et al. (2012)[25]	Probe	0	0	0	Not mentioned	Not mentioned	Intraoral radiographs (radiovisiography)	0
Grandi et al. (2013)[26]	Vestibular pictures including the two adjacent teeth	Not mentioned	N=6 score 1 (difference between 0-1) N=3 score 2 (difference between1-2) (*n*=23)	4	N=4 Jemt score 2 (at least one-half of the height); N=19 score 3 (complete closure of the proximal space)	Not mentioned	Radiographs were scanned and digitized	0.35±0.23 (*n*=23)
Cooper et al. (2014)[3]	UNC -15 probe	0	0.06± 0.98	Mesial = 0.13±1.61; Distal=0.21 ±1.61 (*n*=44)	No	Not mentioned	Periapical radiographs	0.43 ± 0.63
Noelken et al. (2015)[27]	Not mentioned	Not mentioned	0	0	PES	PES= 11.9 ± 1.4	Digital periapical radiographs; Cone beam computed tomography for facial bone	Interproximal. = −0.2 ±0.4 (*n*=18) CBCT of facial bone 1mm from crest = 1.5 ±0.9 3mm from crest = 1.3 ±0.9 6mm from crest =1.4 ±1.0 (*n*=11)
Ma et al. (2019)[28]	Cast using digital caliper	Not mentioned	0.08± 0.20	0	Jemt's classification Mesial papilla N=2 index1; N=8 index2; N=7 index 3 Distal papilla N=4 index1;N=7 index2; N=6 index 3	Not mentioned	Intraoral radiographs	0.1 ± 0.25
Mangano et al. (2013)[29]	Intraoral photographs	Not mentioned	7% cases showed >1mm	Mesial = 1– 8 (score 2 –14); Distal =1–9 (score 2–13)	PES	Total=14.50; PES=7.45; WES=7.04	Intraoral periapical radiographs	Not mentioned
Guarnieri et al. (2014)[30]	UNC 15 probe	0	0.12; 0.1 (found in 2/44 ;4.5% cases)	Mesial = 0.1 (1/ 44; 2.7%); Distal=0.1 (1/ 44;2.7%)	PES	Total =21.3; PES=12.25; WES=8.81	Tomograms and periapical radiographs	Mesial = − 0.58 Distal= − 0.57
Crespi et al. (2008)[31]	Not mentioned	Not mentioned	Not mentioned	Not mentioned	Not mentioned	Not mentioned	Intraoral digital periapical radiographs	Mesial= 0.93 ± 0.51; Distal=1.1 ± 0.27
Block et al. (2009)[32]	The distances from the pre-extraction measurement stent to the gingival margin and papilla on the direct buccal, as well as the distance from the stent to the papilla (at the line angles) using probe	Not mentioned	0	Not mentioned	Papilla score according to Jemt	Not mentioned	Digital periapical radiographs.	Implant to bone =0.45787±1.22019; Cementoenamel junction =2.8204±1.14805; Horizontal=3.03062±0.98772

Table 1. continued

Reference	Measurement method (soft tissue)	Probing depth (>3mm)	Mid-facial recession (mm)	Papillary recession (mm)	Papillary index used	Pink esthetic score(PES)/ white esthetic score (WES)	Measurement method (hard tissue)	Bone loss (mm or %)
Canullo et al. (2010)[33]	Probe	0	Not mentioned	Not mentioned	Not mentioned	Not mentioned	Digital standardised periapical radiographs, computerised measuring technique and digital subtraction radiography (DSR) software	Provisional abutment = 0.55 Definitive abutment = 0.34
Pieri et al. (2011)[34]	UNC-15 probe	0	Test group =0.61±0.54; Control group = 0.73 ± 0.52	Test group: Mesial = 0.24±0.21; Distal =0.28±0.19 Control group: Mesial= 0.33±0.19 Distal = 0.33±0.23	Not mentioned	Not mentioned	Standardized periapical radiographs	Test =0.2±0.17 Control=0.51±0.24
Spinato et al. (2012)[35]	Periodontal probe	Grafted sites= 3.6 (mesiobuccally and distobuccally) (*n*=22) Non grafted= 3.2 mesiobuccally and 3.5 distobuccally (*n*=23)	Grafted sites = 0.40±0.60 Non grafted= 0.30±0.36	Score 2 -3 in 95% of implants in both grafted and non-grafted sites	Papilla index score (PIS)	Not mentioned	Periapical radiographs	Grafted sites = - 0.94± 0.51; Non- grafted site = -0.90 ± 0.49
Esposito et al. (2015)[36]	Not mentioned	Not mentioned	Not mentioned	Not mentioned	PES	PES=13.0	Intraoral periapical radiographs	0.13±0.16 (*n*=52) (1 year)
Felice et al. (2015)[37]	Not mentioned	Not mentioned	Not mentioned	Not mentioned	PES	PES=12.78	Intraoral periapical radiographs	0.13±0.09 (*n*=23)
Crespi et al. (2019)[38]	Pressure sensitive probe (0.35N), periodontal probe	Not mentioned	-0.30±0.58	Not mentioned	Not mentioned	Not mentioned	Not mentioned	Not mentioned
Lee et al. (2020)[39]	Periodontal probe	Not mentioned	0.32±0.44 (*n*=34)	0.17±0.29 (*n*=34)	Not mentioned	Not mentioned	Digital intraoral periapical radiographs	0.81±0.90 (*n*=36)

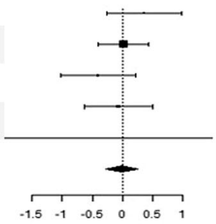
Reference	Mesial papilla recession			Distal papilla recession			Weight	Mean difference IV, Fixed, 95%CI
	Mean	SD	Total	Mean	SD	Total		
Cooper et al. (2014)	0.13	1.61	48	0.21	1.61	48	37.43%	-0.05[-0.45, 0.35]
Guarnieri et al. (2014)	0.04	0.58	44	0.03	0.83	44	34.32%	0.01[-0.40, 0.43]
Pieri et al. (2011)	0.28	0.04	38	0.30	0.02	38	28.25%	-0.63[-1.09,-0.17]
Total (95% CI)			**130**			**130**	**100%**	**-0.19[-0.44, 0.05]**



Heterogeneity: $Tau^2 = 0$; $Chi^2 = 4.83$; df = 2 (P=0.089); $I^2 = 58.61\%$
Test for overall effect: Z = -1.53 (P=0.12)

Fig. 2 Papillary loss.

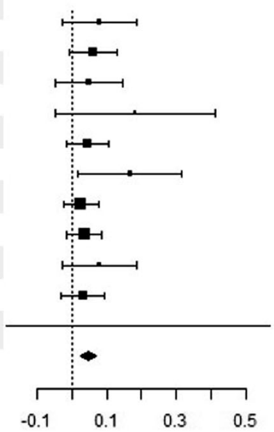
Reference	Mesial bone loss			Distal bone loss			Weight	Mean difference IV, Fixed, 95%CI
	Mean	SD	Total	Mean	SD	Total		
Noelken et al. (2015)	-0.1	0.5	20	- 0.3	0.6	20	18.36%	0.35[-0.27,0.98]
Guarnieri et al. (2014)	0.58	0.53	44	0.57	0.70	44	41.02%	0.02[-0.40,0.43]
Crespi et al. (2008)	0.93	0.51	20	1.1	0.27	20	18.26%	-0.41[-1.03,0.22]
Block et al. (2009)	0.09	1.26	24	0.17	1.13	24	22.36%	-0.07[-0.63,0.50]
Total (95% CI)			**108**			**108**	**100%**	**-0.02[-0.29,0.25]**



Heterogeneity: $Tau^2 = 0.00$; Chi^2 =2.91; df = 3 (P =0.40); $I^2 = 0\%$
Test for overall effect: Z = -0.12 (P =0.89)

Fig. 3 Bone loss.

Reference	Implants available at the end of follow up	Implants failed	Weight	Risk estimate IV, Fixed, 95%CI
Grandi et al. (2013)	25	2	4.74%	0.08[-0.03, 0.19]
Cooper et al. (2014)	48	3	11.42%	0.06[-0.01, 0.13]
Noelken et al. (2015)	20	1	5.87%	0.05[-0.05, 0.15]
Ma et al. (2019)	11	2	1.03%	0.18[-0.05, 0.41]
Guarnieri et al. (2014)	44	2	14.14%	0.05[-0.02, 0.11]
Block et al. (2009)	24	4	2.41%	0.17[0.02, 0.32]
Pieri et al. (2011)	38	1	20.68%	0.03[-0.02, 0.08]
Esposito et al. (2015)	54	2	21.12%	0.04[-0.01, 0.09]
Felice et al. (2015)	25	2	4.74%	0.08[-0.03, 0.19]
Lee et al. (2020)	31	1	13.85%	0.03[-0.03, 0.09]
Total (95% CI)	**487**	**20**	**100%**	**0.05[0.02, 0.07]**



Heterogeneity: $Tau^2 = 0$; Chi^2 =5.75; df = 9 (P=0.764); $I^2 = 0\%$
Test for overall effect: Z =4.04 (P<0.001)

Fig. 4 Implant loss.

5 studies[27,29,30,36,37]. Hard tissue assessment by intraoral periapical radiographs was done by 9 studies[3,26,28–30,34–37], digital radiographs by 6 studies[25,27,31–33,39], CBCT by 2 studies[27,30] of which 13 studies[3,26–28,30–37,39] reported on bone loss.

Quality assessment

The quality assessment of the 7 prospective and retrospective cohort studies[3,25–30] was performed according to the NOS[23], and the scores obtained are shown in Supplementary Table 3. All the 7 studies were of high quality. The Cochrane collaboration's tool[24] was used to assess the risk of bias for 9 randomised controlled trials[31–39] and the scores are summarised in Supplementary Table 4, all studies were found to have low risk of bias.

Meta-analysis

Included studies showed moderate heterogeneity ($I^2 = 58.61\%$; $P = 0.089$) in the forest plot for the event papillary recession (Fig. 2), the Cochran's Q was 4.83 and was not significant ($P = 0.12$). Included studies showed no heterogeneity ($I^2 = 0\%$; $P = 0.40$) in the forest plot for the event bone loss (Fig. 3) and the Cochran's Q was 2.91 and was not significant ($P = 0.89$). Included studies showed no heterogeneity ($I^2 = 0\%$; $P = 0.764$) in the forest plot for the event implant failure (Fig. 4), the Cochran's Q was 5.75 and was highly significant ($P < 0.001$).

Publication bias

The assessment of publication bias is presented in Supplementary Table 5. The funnel plots for the events papillary recession (Supplementary Fig 1), bone loss (Supplementary Fig 2) and implant failure (Supplementary Fig 3) showed symmetry, suggesting presence of no publication bias.

DISCUSSION

According to the Third ITI Consensus Conference, the placement of implants is categorized based on the timing of healing after tooth extraction. This includes Type 1 immediate (within 24 h), Type 2 early (4–8 weeks), Type 3 early-delayed (12–16 weeks), and Type 4 late (more than 6 months). The classification takes into account the morphologic, dimensional, and histologic changes of healing sockets to approximate the soft and hard tissue characteristics[40]. In this systematic review, the soft and hard tissue changes and success/ survival rate of Type 1 placements were subjected to evaluation.

Flap vs flapless approach

During the surgical procedure for IPI placement, there is a choice between making an incision and detachment or opting for a flapless surgery. When using an open technique, the blood supply from the periosteum is restricted. By not lifting the flap for implant insertion and performing the surgery through the alveolus, the vascularization can be preserved, resulting in improved post-operative recovery and healing.

Among the included studies in the present systematic review, Cooper et al.[3] found minor difference in two groups where implants were placed using a flapped or flapless procedure at the 5-year evaluation. It is generally accepted that flapless procedures offer aesthetic advantages, but few comparative studies like Bashutski et al.[41] reported no difference between flapped and flapless approaches in a 15-months follow-up. Implants placed into healed ridges using a flapless technique revealed a slight shift in the peri-implant mucosal position after 5 years, but not around implants inserted into extraction sockets. The variations in mucosal zenith stability between the healed ridge group and extraction sockets may be attributed to buccal bone resorption and ongoing bone and connective tissue remodelling. The long-term implications of these minor alterations remain uncertain, especially considering other factors like adjacent tooth extrusion near single-tooth implants[3].

Block et al.[32] compared and found there was no significant correlation between the use of a flap versus no flap and the movement of the facial gingival margin and papilla height over time or at baseline ($P > 0.05$). Additionally, the use of a flap to expose the bone in the delayed group did not impact the final position of the papilla.

Lee et al.[39] reported more interproximal gingival recession in flap elevation than flapless approach, which is due to more crestal bone loss caused by flap elevation, although the difference in crestal bone loss was not statistically significant between the flap-involving and flapless groups. Impact of age and reduced blood flow in aged gingiva may also affect the stability of the interproximal gingiva but their clinical impact might not be considerable.

Grafted vs non-grafted sites

The occurrence of GAP is discussed by many authors as a frequent scenario during the IPI, sparking a debate on the necessity of grafting or not using autogenous bone, alloplastic materials, or membranes[1].

Factors influencing the resorption of the buccal bone wall in horizontal dimensions after immediate implant insertion using a flap procedure and no defect grafting have been reported by prospective studies[42,43]. To prevent the facial bone wall resorption and to enable reconstruction in the study by Noelken et al.[27] the implants were placed palatally by flapless approach, and the horizontal gap on facial aspect was filled with autogenous bone chips. Canullo et al.[33] reported when the gap between implant and inner aspect of the neighbouring bone was more than 1 mm, it was grafted with nano-structured hydroxyapatite and covered with a collagen disc to maintain the graft material in situ and to prevent particle leakage, this procedure could improve the aesthetic outcome. Chen et al.[44] found no statistically significant differences in the trial comparing 14 patients who received particulate autogenous bone versus 12 patients who were not subjected to any augmentation procedures in cases of immediate single implant placement in fresh extraction sockets in the maxillary anterior zone. However, both failures and complications were reported in the augmented group suggesting that the augmentation procedure is not free from complications in the postoperative period, probably due to the reduced blood supply by the grafting materials.

It has been reported that the marginal gap between the buccal bone and the implant surface in an extraction socket heals predictably with new bone formation and defect resolution without grafting materials[45,46] but this depends on both the thickness of the buccal bone wall and the size of the horizontal gap[47].

Paolantonio et al.[42] demonstrated that a horizontal gap of ≤2 mm was resolved through new bone formation spontaneously; and another study demonstrated that sites with horizontal gap of >1 mm the degree of gap fill was substantial[47]. In the study by Spinato et al.[35] the average dimension of the horizontal defect was 2.14 mm (range 1 to 3.5 mm) which is in agreement with the above authors because the 23 no graft sites showed complete bone formation without any complications. There are contradicting reports in the literature[2] regarding the influence of the initial bone fill amount on the final bone fill especially in the horizontal gap which needs to be investigated.

Lee et al.[39] found that wider HDD (> 1 mm) of the immediate implant sites were associated with a greater buccal ridge reduction as it prevents the risk of biologic dimension violation. Greater bone reduction is usually seen in a wide and not grafted horizontal defect because of an increased risk of the blood clot dislodgement. Non-grafted immediate implant sites showed stable gingival level and ridge dimension with less than 1 mm change during the 12-month follow-up period which is in accordance with various studies[48,49].

However, it is known that bone grafts or the connective tissue graft can be used in the immediate implant site to preserve or augment tissue dimensions. Reduction of ridge dimension or gingival recession in the immediate implant sites with bone grafts could be 0.5 to 1 mm less than in the non-grafted sites[48–50]. Stable gingival level, increase gingival thickness, and preservation of ridge dimension can be achieved by placing the connective tissue graft at the immediate implant site[50–52]. Hence, according to the included studies and literature, use of bone grafts or the connective tissue graft in the immediate implant site is not required for every case but it is recommended when minimal tissue alterations are anticipated[39].

Gingival biotype: thick vs thin

According to Muller et al.[53] patients' tissue biotype was evaluated as either thin or thick based on the specified criteria:

1. In thin tissue, a periodontal probe placed into the labial gingival sulcus is visible through the gingival tissue.
2. In thick tissue, a periodontal probe placed into the labial gingival sulcus is not visible through the gingival tissue.

The form and biotype of the periodontium play a crucial role in achieving the best aesthetic results when placing an immediate implant[54–56]. A thick periodontal biotype results in an increased peri-implant mucosal dimension, providing greater resistance to gingival recession[57], hence, patients with thick biotypes are more suitable for immediate implant placement due to the reduced risk of tissue recession after the procedure, leading to consistent aesthetic outcomes[55,56]. Placing immediate implants in patients with a thin biotype may lead to an increased chance of soft tissue recession and resorptive osseous remodelling, which can result in the exposure of the implant's metal margin[54,56]. A thick, flat periodontium is characterized by relatively flat bony architecture and a thick osseous form that is resistant to vertical resorption and may increase the amount of horizontal gap fill[47,58].

In contrast to the previously mentioned findings, Lee et al.[39] observed that a thick gingival phenotype was linked to greater mid-buccal gingival recession. The authors explained that this might be due to baseline submarginal oedema, which could have affected the gingival thickness measurements, as fractured and heavily decayed teeth were included in the study. Consequently, areas classified as "thick phenotype" may have been influenced by inflammation that reduced after tooth extraction. They also noted that a thick gingival margin phenotype was associated with less ridge dimensional reduction at the 12-month follow-up, likely due to the extensive supracrestal connective tissue attachment in thick phenotypes, which may help prevent bone resorption as the peri-implant mucosal dimension forms around the implant[59,60].

Guarnieri et al.[30] concluded that clinical experience and careful case selection of patients with a thick gingival biotype, ideal gingival level/contour, and intact socket walls at the time of tooth extraction are deemed mandatory for the success of this kind of treatment strategy.

Torque values and immediate implant placement

Primary stability is a biomechanical aspect to be considered for immediate implant loading[61,62]. Various experimental studies have stated that micromotion of implants should not exceed a threshold of 50 to 150 µm; otherwise, fibrous encapsulation of implants takes place, rather than osseointegration[63,64]. Hence, high primary stability is necessary for immediate loading of dental implants; for this reason, modified drilling protocols in combination with bone compaction with osteotomes have been proposed[65,66].

Some authors have chosen insertion torque as a measure of implant stability, but no consensus has been achieved concerning a torque threshold value[11,67]. In addition, successful immediate loading of an implant with an insertion torque of just 15 Ncm has been accomplished under some conditions[66]. To date, no controlled study has compared the relationship of different implant stability values and implant survival rate; as a consequence, there is currently no proven threshold value above which immediate loading is ensured to be successful[25].

In the study by Grandi et al.[26] implants inserted with a torque of at least 45 Ncm underwent an immediate non-occlusal loading protocol. Additionally, the torque values for implant insertion were notably higher in fresh extraction sockets compared to delayed implants. The data suggest that even after 4 months of healing, the Bio-Oss-filled alveoli remained soft, hence obtaining high insertion torque was very difficult compared to immediate post-extraction sites[68].

Felice et al.[37] observed that it was not always possible to achieve an insertion torque of at least 35 Ncm at implant placement, especially in delayed implants sites due to similar reasons as mentioned above.

As per Osstell's recommendation, an implant stability quotient (ISQ) value of over 70 indicates high stability, while a value between 60–69 is classified as medium stability and any ISQ value below 60 is categorized as low stability[69,70]. Ma et al.[28] used resonance frequency analysis for the measurement of implant stability for participants at 5 years recall, which ranged from ISQ value of 58 (at time of implant placement) to 78 (Year 5 recall), these changes in the ISQ values is expected as there is ongoing healing and remodelling at the implant site. This also shows that the primary stability achieved at the time of immediate implant placement not only improved during the first year of function but also successfully maintained up to 5 years. Esposito et al.[36] reported that implants placed with a minimum torque of 35 Ncm were immediately loaded with temporary non-functional resin crowns. Patients in the immediate group could come in with a tooth scheduled for extraction and leave on the same day with a visually pleasing implant-supported crown.

Immediate implant placement vs early &/or delayed implant placement

Various included studies[3,26,29,32] reported no differences in complications or statistically significant differences in crestal bone response at immediate post-extractive implants immediately restored versus delayed implants immediately temporised in the anterior maxilla.

Throughout this time frame, the marginal bone levels and soft tissue levels maintained a close proximity to the implant/abutment interface and the prosthetic reference (mucosal zenith), respectively. The tissue architecture remained stable during the entire study period. Authors concluded that immediate implant placement may not show an increased risk for aesthetic failure when treatment is performed by experienced clinicians in well-selected cases.

Esposito et al.[36] and Felice et al.[37] concluded that there were immediate post-extractive implants present greater challenges compared to delayed implants. It appears to be more challenging to achieve an implant insertion torque exceeding 35 Ncm in sockets preserved with bone substitute after a 4-month healing period than in post-extractive sites. However, the aesthetic results seemed to be similar for both the groups.

Esposito et al.[6] in a review concluded that limited data makes it difficult to assess the potential pros and cons of immediate, immediate-delayed, or delayed implants. These initial findings rely on a small number of inadequately powered trials with a high risk of bias. Some evidence suggests that immediate and immediate-delayed implants may be more prone to failures and complications compared to delayed implants[71]. However, there is a possibility of achieving a better aesthetic outcome by placing implants immediately after tooth extraction.

Immediate loading versus delayed loading

In the study by Crespi et al.[31] on 2 years follow-up found no significant clinical or radiographic differences between immediate and delayed loading of implants placed in fresh extraction sockets, indicating that immediate restoration of single-tooth implants in these sockets can be a reliable approach that boosts patient satisfaction.

Another study[38] by same author compared midfacial soft tissue assessment and horizontal width changes of fresh socket implants in immediate and delayed prosthetic restorations, and at 4 years follow up found that immediate implants restored on the day of surgery demonstrated more stable mid-facial soft tissue levels and bone volume maintenance than delayed implants.

But when it comes to comparison between immediate functional and non-functional loading of implants in fresh extraction sockets, to date, there are no controlled studies that provide sufficient evidence to determine whether single-tooth implants should be loaded immediately or restored without occlusal contacts.

Aesthetics outcomes following immediate implant placement and loading

Bone resorption is inevitable following tooth extraction accompanied by soft tissue recession[72–74]. and this affects the aesthetic outcome of implants placed in aesthetic areas, especially in patients with high smile line. Two methods are advocated when planning an implant restoration in an extraction site in order to prevent or minimise the incidence and amount of hard and soft tissue grafting. One approach involves grafting the extraction site with a bone substitute, while the other entails immediate implant placement. Immediate post-extraction implant placement is widely recognized for its aesthetic benefits, reduced treatment duration, and enhanced patient comfort, with well documented long-term outcomes of immediate implant placement and provisional restoration[75].

Various indices have been proposed for assessment of papilla fill. In 1997, Jemt[57] proposed an index (papilla filled index) for assessing the size of the interproximal gingival papilla. Fu¨rhauser et al.[76] proposed The pink aesthetic score (PES) is utilized to objectively assess peri-implant soft tissue aesthetics. The PES concept primarily emphasizes the soft tissue elements of an anterior implant restoration. Nonetheless, additional factors contribute to the overall aesthetic result of implant-supported single-tooth replacement, including variables related to the crown. An aesthetic implant restoration should closely resemble a natural tooth in every aspect. To achieve this goal, Meijer et al.[77] published he aesthetic implant-crown index has been released, which includes criteria pertaining to both the implant restoration and the soft tissues surrounding it. Belser et al.[78] recently introduced the pink aesthetic/white aesthetic score (PES/WES), which not only focuses on the soft tissue aspects of an anterior implant restoration but also on the visible part of the implant restoration itself. "The PES/WES includes 10 variables: mesial papilla, distal papilla, curvature of the facial mucosa, level of the facial mucosa, and root convexity/soft tissue colour and texture at the facial aspect of the implant site, tooth form, volume, colour, surface texture, and translucency. A score of 2, 1, or 0 is assigned to all parameters, which are assessed by direct comparison with the natural, contralateral reference tooth to estimate the degree of match or eventual mismatch."[78] Cho et al. verified the appropriateness of the PES/WES index for the objective evaluation of the aesthetic aspect of anterior single-tooth implants[79].

Di Alberti et al.[25] reported that interproximal papillae were preserved at all sites, and no signs of soft tissue recession or other defects were visible. The peri-implant tissue responses at the midfacial and interproximal areas of implants inserted in both healed ridges and sockets was evaluated by Cooper et al.[3] Here, favourable responses were seen in both groups, with no midfacial resorption and stable papilla length measured from baseline. Authors used the PES-WES to evaluate peri-implant and implant crown aesthetics. Noelken et al.[27] reported a stable facial soft tissue level in the PES (preop 1.63, 3-year follow-up 1.72). Similar results were found by Ma et al.[28] between at Year 1 and Year 5 follow-up. In study by Mangano et al.[29] the mean PES/WES aesthetic outcome was 14.50 (SD 2.52; range 9–19) and 15.61 (SD 3.20; range 8–20) for immediately and conventionally placed implants, respectively. No significant differences were found between the 2 groups. Guarnieri et al.[30] reported that mesial and distal papillae demonstrated a significant regrowth from baseline to 2 years, with mesial papillae showing an average regrowth of 1.8 mm and distal papillae with an average regrowth of 1.5 mm. The midfacial soft tissue levels remained relatively stable over time, with a mean recession of 0.12 mm compared to the preoperative status. The PES/WES results from this study revealed that 30 out of 44 cases (68%) achieved an almost perfect result ($PES \geq 12$ and $WES \geq 9$). The stringent selection inclusion criteria could be attributed to the favourable aesthetic outcomes observed.

Block et al.[32] found that the facial gingival margin, quantified as facial gingival height from the reference point, were different when implants were placed. The immediate group demonstrated a statistically significant more coronally positioned facial gingival margin in comparison to the delayed group. In the delayed group, the tooth extraction and socket graft placement led to an approximate 1 mm greater recession of the facial gingival margin, whereas the immediate group benefited from support provided by the provisional crown. The choice of group did not impact the levels of the papilla.

Pieri et al.[34] found the position of the soft tissues, reflected in the parameters MFL, MPL, and DPL, changed significantly between baseline and 4 months for both groups, whereas no significant changes were detected from 4 to 12 months. In all patients, most of the recession occurred during the first 4 months of loading, but no significant variation was detected between 4 and 12 months, which is indicative of a steady state. This magnitude of soft tissue recession during the early phase of functional loading is in accordance with previously published data[80,81]. With regard to the dimensions of the mesial and distal papillae, the mean amount of recession after 1 year was minimal in both the control and test groups and remained virtually stable during the whole period of the study.

Spinato et al.[35] reported the interproximal papillae between implant and adjacent teeth were stable over time with papillae present in 95% of implants placed and no difference was found between two groups. Apical movement of the gingival contour (i.e, gingival recession) is a common occurrence after implant restorations[55] and approximately 1 mm of tissue recession can be expected[81–83]. Nevertheless, during the 1-year follow-up appointment, Kan et al.[84] documented a decrease in recession (0.55 mm). These slight changes in gingival tissue suggest that the peri-implant tissue remained stable over time. There was only one instance of a midfacial probing depth of 7 mm, which was attributed to incomplete bone fill in the grafted group.

Esposito et al.[36] found that one year after loading, the average PES score was 13.0 for the immediate group and 12.8 for the delayed group, the difference showing no statistical significance. This could be interpreted as both procedures achieving the same aesthetic outcome. Felice et al.[37] also found similar results 1 year after loading. Crespi et al.[38] in the study of 4-years follow-up showed significant differences ($P < 0.001$) between the groups. Immediate implant placement group showed better stability of midfacial soft tissue levels and bone volume retention compared to delayed implants. Lee et al.[39] concluded that gingival phenotype is a key factor influencing changes at the mid-buccal gingival margin and mid-buccal ridge dimension in immediate implant placement sites whereas increased interproximal gingival recession was associated following flap elevation.

Contradictory results regarding the aesthetic outcome were reported by Grandi et al.[26] where the immediate group showed a significantly higher gingival recession. An ideal gingival marginal level was found in 52.1% of the immediate group compared to 83.3% of the delayed group. The improved clinical marginal buccal gingival level in the delayed group could be attributed to the increase in attached gingiva following full mucosal wound healing[85]. The immediate group showed a higher papilla score, but the disparity in score prevalence between groups did not reach statistical significance. Immediate implant placement may be beneficial in maintaining the integrity of extraction sockets and also help in the maintenance of interdental papillae around implant restorations[85,86].

Di Alberti et al.[25] concluded that restoration of the aesthetic area with implant supported prostheses is one of the most difficult and challenging procedures to execute, since bone resorption patterns, implant positioning, prosthetic provisionalization, and definitive restorations are all key factors in treatment success. It is known that when a single maxillary tooth is missing, the buccal bone loses volume, while the mesial and distal bone crests remain in place. Following this theory, palatal and central positioning of single implants was performed with success. Therefore, the presence of the interproximal bone peaks and interdental papillae and their preservation as a result of additional adjustments of the provisional crowns may lead to the preservation of hard and soft tissue around a complete restoration in the aesthetic zone.

Bone-level changes following immediate implant placement and loading

Di Alberti et al.[25] found that periapical radiographs showed no signs of infrabony defects or marginal bone loss. Comparison of the most recent radiographs to the previous ones revealed that the proximal peaks had been preserved in immediate implant placement and loading. Cooper et al.[3] found that, over the 5-year evaluation period, the interproximal bone levels at implants placed in sockets and healed ridges attained a position within 0.5 mm of the established reference point. Ma et al.[28] revealed that the marginal bone loss observed was negligible and remained consistent from year 1 to year 5 (average loss = 0.1 mm, standard deviation 0.25). Those individuals who completed the 5 years follow-up were classified as successful since the marginal bone level alterations were below 1.8 mm throughout the 5-year period. Crespi et al.[31] showed there were no statistically significant variances ($P > 0.05$) observed in mesial or distal crestal bone loss between the delayed and immediate loading groups. The immediate loading group exhibited lower bone loss, which could be attributed to the response of the peri-implant bone to immediate loading. This is supported by previous studies where immediately loaded implants showed significantly higher bone-implant contact compared to implants placed using a 2-stage protocol[87,88].

Grandi et al.[26]. reported that no significant difference in radiographic bone level was seen between the immediate and delayed groups during the first year of function. Immediate implant placement post-tooth extraction did not result in superior preservation of crestal bone compared to the delayed approach. After 12 months, patients in the immediate group experienced an average loss of 0.71 mm of peri-implant bone, while those in the delayed group lost 0.60 mm. Block et al.[32] discovered that the bone to implant crestal bone level shifted apically after abutment placement, aligning with the abutment implant interface. The level of the adjacent bone remained constant over time. No disparities were observed when comparing different tooth locations from premolar to incisor. This suggests that the ultimate bone crestal levels will be influenced by the depth and configuration of the implant, indicating that immediate implant placement post-tooth extraction did not yield greater preservation of crestal bone compared to a delayed approach. Spinato et al.[35] reported peri-implant bone loss was slightly more pronounced (0.65 mm at T1 and 0.94 mm at T2 for grafted sites and 0.55 mm at T1 and 0.90 mm at T2 for non-grafted sites) compared to previous studies[83,84]. Esposito et al.[36] found significantly more bone loss for delayed implants (0.27 mm) than immediate implants (0.13 mm), and stated that a difference of 0.14 mm between the two groups cannot be considered clinically relevant since it was not noticeable, which is similar to the findings of Felice et al[37].

Lee et al.[39] reported that no significant correlation was found between the examined factors with the change in the interproximal marginal or crestal bone level after 12 months. It is worth noting that the average 12-month marginal bone level (bone-implant contact) at the sites where immediate implants were placed (2.37 ± 1.20 mm) was typically 1 mm more apical compared to the marginal bone level at sites where delayed implants were inserted. Since no bone graft was employed in this particular study, this outcome might suggest a restriction in bone formation around immediately placed implants without additional regenerative procedures.

According to the review by Lang et al.[89] in the initial 3 months after immediate implant placement and restoration, most of the marginal bone loss occurred, while by the end of the first year, the bone loss was typically less than 1 mm. One potential method to reduce hard tissue changes in the short term is through the use of platform-switching technique, which involves restoring a wider-diameter implant with a narrower-diameter abutment. A randomized clinical controlled trial showed significantly less bone resorption near platform-switched abutment restorations compared to platform-matched abutments[90]. However, another RCT did not demonstrate such differences[91]. Therefore, further clinical trials are necessary to validate the potential advantages of the platform-switching technique.

Over time, proper oral hygiene is very essential for maintaining bone levels. Bone levels showed improvement (average gain of 0.2 mm) after 5 years of implant use, particularly with low plaque and mucositis levels[92].

Implant failures, success and survival rates following immediate implant placement and loading

Over the past few years, there has been a rise in the amount of research focusing on immediate loading, with many studies reporting a high success rate for implant treatments in their findings[1]. Studies centred around implant survival rates when it comes to immediate implant placement and loading plays a crucial role in determining the success of the treatment[30]

Di Alberti et al.[25] found that at the 1-year follow-up appointment, no implants had been lost, and all the implant-supported single crowns were clinically stable and the success rate after 12 months of follow-up was 100%. In the study by Grandi et al.[26] two implants failed in the immediate implant group with survival rate of 92% versus 100% for delayed implants. Cooper et al.[3] reported the overall implant survival rate of 96.5%. Spinato et al. The cumulative implant survival rate was 100% after at least 1 year of function with an average follow-up period of 32 months for immediate placement and provisionalization without flap elevation which are in accordance to (two articles11,31). Similarly, a comparable success rate (>97%) has been reported using the same protocol but with full-thickness flap elevation.32–34. Noelken et al.[27] reported a success rate of 90% according to the Kaplan–Meier method.

Mangano et al.[29] reported that with regard to osseointegration, all 40 anterior maxillary single-tooth Morse taper connection implants met the success criteria for osseointegration, achieving a 100% success rate for implant-crown. Throughout the first year of function, all implants demonstrated stable osseointegration, without any pain, suppuration, clinically detectable implant mobility, peri-implant radiolucency, DIB exceeding 1.5 mm, or prosthetic complications at the implant-abutment interface.

Guarnieri et al.[30] reported that after 2 years of loading, 95.6% of the implants survived. A meta-analysis from a recent literature review also reported a comparable overall survival rate (95.5%) of implants in the aesthetic zone within a short time frame[20]. Crespi et al.[31] reported a survival rate of 100% for all implants after a 24-month follow-up period. Pieri et al.[34] reported implant success was 100% in the control group and 94.7% in the test group.

Spinato et al.[35] reported the immediate placement and provisionalization procedure in the present study resulted in a 100% (45/45) cumulative implant survival rate after a minimum of 1 year of function, with an average follow-up period of 32 months. Only two articles[11,31] in the literature reported a similar survival rate (100%) without flap elevation. Another study using the same protocol but with full-thickness flap elevation also reported a comparable success rate (≥ 97%)[32–34].

Esposito et al.[36] reported two implants failure from the immediate group, which means that 4% of the immediate post-extractive implants failed, leading to success rate of 96%. In the study by Felice et al.[37] two implants (8%) failed in the immediate post-extraction group, a rate that aligns with expectations. These results reinforce the idea that immediately-loaded implants placed after extraction may carry a greater risk of failure compared to delayed implants, consistent with findings from other RCTs testing this hypothesis.

CONCLUSION

In conclusion, among the studies included in the present systematic review and meta-analysis:

- No differences were found in crestal bone levels when comparing immediate or delayed implant placement with immediate provisionalization in the anterior maxilla.
- The peri-implant margin remained stable in most of the included studies and no differences in papillary loss was seen when compared to the delayed implant placement cases.
- Papilla were more stable or showed less recession in flapless approach compared to full thickness flap approach.
- Among the studies which filled the GAP with bone graft materials, no significant changes were found in the bone level changes. And when it comes to recession, immediate implant placement with provisionalization did result in approximately 1 mm less facial gingival recession compared with that in the group that had a socket graft.
- Implant related complications occurred more in immediate implant placement and provisionalization compared to delayed group.
- Almost similar implant success and survival rates were seen in comparison to delayed implant placement groups.

The findings of this study add onto the prior information on the outcomes of immediate implant placement and loading that the aesthetic and hard tissue outcome appears to be similar for both groups thereby proving that implant placement in fresh sockets did not prevent tissue contraction, and demonstrated more stable midfacial soft tissue levels, papillary height and bone volume maintenance than delayed implant placement.

DATA AVAILABILITY

The authors confirm that the data supporting the findings within this study are available within this article.

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AUTHOR CONTRIBUTIONS

RG reviewed the existing literature evidence base. RG, RC, SM, KS constructed the search strategy. The designed search strategy was approved and validated by all authors. RG and KS conducted and completed the article selection in accordance with delineated inclusion and exclusion criteria. SM and RC re-assessed the article selection. RG and KS conducted the extraction of data. RC and SM re-examined and validated the data. RG and RC conducted the quality assessments and risk of bias of included studies. RG and SM conducted the meta-analysis. RG, RC, SM and KS analyzed all the results. RG wrote the first draft of the article and prepared the tables and figures. All authors reviewed and approved the final version of the manuscript.

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COMPETING INTERESTS

The authors declare no competing interests.

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